10TH CLASS COMPUTER SCIENCE — NEW SYLLABUS (PECTAA)

Chapter 1 — Operating Systems

Structure and Services | Solved Exercise | New Book (PECTAA)

Exercise (Solved)

A. Multiple Choice Questions

1. Which is NOT an example of an operating system?

(a) Linux (b) Windows
(c) Photoshop (d) macOS

Answer: (c) — Photoshop is an application software used for editing pictures. Linux, Windows and macOS are operating systems, that is, system software which manages the hardware and runs the application software.

2. In multi-user environments, the OS ensures:

(a) Equal access and data privacy (b) All users share the same files
(c) Hardware is never used (d) Internet speed is increased

Answer: (a) — In a multi-user environment the operating system creates and manages separate user accounts, keeps each user's files and information private, shares the computer's resources fairly and protects the shared resources from unauthorised access.

3. In Windows, user accounts can be created via:

(a) Task Manager (b) Control Panel or Settings
(c) Disk Management (d) BIOS settings

Answer: (b) — In Windows a user account is created through Settings or Control Panel by opening User Accounts, choosing Add New User and entering the username, password and account type (Standard, Administrative or Guest). Task Manager and Disk Management do not create accounts.

4. The core part of the OS that interacts directly with hardware is the:

(a) Shell (b) Kernel
(c) Device Driver (d) System Library

Answer: (b) — The kernel is the core part of the operating system which directly controls the hardware and the system software such as the CPU, memory and devices. The shell is the outer part that talks to the user.

5. A graphical shell allows the user to:

(a) Type commands only (b) Click icons and use menus
(c) Interact only via code (d) Access only the BIOS

Answer: (b) — Shells are of two kinds. Graphical shells, such as the Windows desktop, allow the user to click icons and use menus. Command-line shells, such as the Command Prompt or Terminal, are used by typing instructions.

6. In the FCFS scheduling method, processes are served:

(a) Randomly
(b) Shortest job first
(c) In the order they arrive
(d) By priority only

Answer: (c) — In First Come, First Served (FCFS) scheduling the CPU processes the tasks in the exact order in which they arrive. The first process to arrive is completed first, and the next process starts only after the previous one finishes.

7. Threads in the same process:

(a) Have separate memory spaces
(b) Share the same memory and resources
(c) Run on different OS
(d) Cannot run simultaneously

Answer: (b) — A thread is the smallest unit of execution within a process. All the threads of the same process share the same memory and resources, but each of them operates independently, which makes execution faster.

8. Which system call is used to create a new process?

(a) Open
(b) read
(c) write
(d) fork

Answer: (d) — The fork system call is used to create a new process. The open, read and write system calls are used for files: open opens a file, read takes data from a file and write sends data to a file.

B. Short Questions

1. Why is the operating system referred to as the “central controller”?

The operating system is called the central controller of the computer because it controls and manages everything that happens inside the system. It decides which task should be done first, how much memory each program will get, and which device such as a printer or a screen will be used and at what time. No program can use the hardware directly; every request has to pass through the operating system. In this way it keeps the whole computer working smoothly and securely, even when many programs and many users are working at the same time.

2. How does the OS act as a translator between the user and hardware?

The computer hardware cannot understand human language directly, and the user cannot give instructions in the machine language of the hardware. The operating system works between the two and translates the actions of the user into instructions that the hardware can understand. For example, when you click an icon or type something, the OS converts that action into the low-level instructions which the processor and the devices can carry out. Because of this translation, people can use the computer easily without learning how the hardware actually works.

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3. Define one way to create a new user account in Windows.

In Windows a new user account is created through the **Settings** (or the Control Panel). The steps are: open Settings and go to **Accounts**, choose **Add New User**, and then enter the details such as the username, the password and the account type. The account type may be Standard, Administrative or Guest. After this, the new user gets his own desktop, files, settings and password, so his data remains separate and private.

4. How is kernel different from shell?

Kernel	Shell
It is the core part of the operating system.	It is the outer part of the operating system.
It works directly with the hardware such as the CPU, memory and devices, and decides how and when different programs use these resources.	It interacts with the user, receives commands from him and passes them to the kernel.
It is not visible to the user; it works in the background.	It is the part the user actually sees and uses, for example the desktop or the Command Prompt.
Example: when you open a file, the kernel brings it from the hard drive and sends it to the screen.	Example: a graphical shell with icons and menus, or a command-line shell such as the Terminal.

5. Give one example of a system library and its purpose.

A system library is a collection of ready-made instructions which programs can use to perform common tasks instead of writing the code themselves. One example is the **printf()** function of the standard library, whose purpose is to show text on the screen. In the same way, a program uses the library when it has to open an image file. Because the library already contains this code, the application program does not have to create its own method for such tasks.

6. Explain one advantage and one disadvantage of FCFS scheduling.

Advantage: FCFS is very simple and easy to implement, because the processes are served in the exact order in which they arrive and no process is skipped. It is also fair, since every process gets its turn.

Disadvantage: a short process may have to wait for a long time if it is queued behind longer processes. This is called the **convoy effect**, and it can reduce the overall efficiency of the system. For example, a process which needs only 2 seconds has to wait until the longer processes before it are finished.

7. Why is virtual memory slower than RAM?

Virtual memory is the part of the computer's storage drive, such as a hard drive, an SSD or an NVMe drive, which the operating system uses as extra RAM when the actual RAM becomes full. It is slower than RAM because the storage drives work at a much lower data transfer speed and access rate than RAM. Moreover, the data has to be moved again and again between the RAM and the drive, and this extra work takes more time. That is why, when several programs are open and virtual memory is being used, the less active programs respond slowly.

8. Define a system call.

A system call is a request made by a program to the operating system to perform a specific task which the program cannot do directly. System calls act as a bridge between the user programs and the kernel, and they allow the applications to use the hardware and the core functions of the operating system safely. For example, when you save a file, the program uses a system call to tell the operating system to write the data on the storage drive.

9. What is the role of a file system?

A file system is the way an operating system stores and organizes files on a storage device such as a hard drive, an SSD or a USB. It decides where each file will be kept, saves its location so that it can be found later, and keeps the data organised, secure and easily retrievable even when thousands of files are present. It also keeps the metadata of the files, such as the name, type, size and the dates of creation and modification. For example, when you save a photo, the file system stores it in the right place and knows exactly where to find it when you open it again.

10. Define multitasking in computer system.

Multitasking is the ability of an operating system which allows more than one program to be open and used by a single user at the same time. The user can switch between the programs whenever he needs; for example, he can listen to music, keep a document open and browse the internet at the same time. This is possible because the CPU switches between the processes so rapidly that the user does not notice any delay.

C. Long Questions

1. Describe the architecture of an operating system, explaining kernel, shell, and layered design with examples.

The architecture of an operating system is the way in which its parts are organised and how they work together. Every part has a special role, and together they make the computer work smoothly, just as the different departments of a school perform their own duties but work together for the smooth working of the school.

(i) Kernel: the kernel is the core part of the operating system. It directly controls the computer's hardware and system software such as the CPU, memory and devices, and it decides how and when different programs will use these resources. When you open a file, the kernel manages the whole process of bringing it from the hard drive and sending it to the screen. The kernel may be compared to the engine of a car.

(ii) Shell: the shell is the outer part of the operating system which interacts with the user. It receives the commands of the user and passes them to the kernel. Shells are of two types: **graphical shells**, such as the Windows desktop, in which you click icons and use menus, and **command-line shells**, such as the Command Prompt or the Terminal, in which you type the instructions. Keeping the same example of the car, the steering wheel and the dashboard are the shell.

(iii) System libraries: these are collections of ready-made instructions which programs use to perform common tasks, such as opening a file or showing text on the screen. Because of them, an application does not have to write this code itself.

(iv) Device drivers: these are special programs which allow the operating system to communicate with the hardware devices such as printers, keyboards and graphics cards. Without the driver of a printer, the OS cannot send the document to it.

(v) Layered design: in this design the operating system is divided into layers, and each layer has a specific job. The **lower layer** works directly with the hardware, such as the storage and the hard drive. The **middle layer** manages memory and storage so that the programs can use them properly. The **upper layer** runs the applications and provides the interface which the user sees on the screen. Every layer depends on the layer below it, and this makes the operating system easy to manage, repair and improve without changing the whole system.

Example: a school can be used to understand this design. The support staff, such as guards and cleaners, work at the base and keep the school ready, like the lower layer which works with the hardware. The administration manages the resources, the schedule and the rules, like the middle layer which manages memory and storage. The teachers and the students use all these arrangements to teach and to learn, like the upper layer which runs the applications and interacts with the user.

2. Explain the process lifecycle in detail with examples.

A process is an independent program which is currently being executed by the computer. The operating system is responsible for managing all the processes, so that each of them gets the resources it needs and none of them disturbs the others. Every process passes through the following three stages in its lifecycle.

(i) Creation: this stage happens when you start any program. The operating system loads the program from the storage into the main memory (RAM) and allocates the resources which it needs, such as memory space and a process number.

(ii) Execution: in this stage the process is actively running and performing its tasks. The CPU executes its instructions, and the process may also wait for some time for input or for a device.

(iii) Termination: the process finishes its task and is closed either by the user or by the system. The operating system then frees the resources of the process so that other processes can use them.

Example — opening a web browser such as Google Chrome: the **creation** begins when the user clicks the browser icon; the operating system loads the program into the memory and allocates the required resources. During **execution** the browser loads the web pages, displays the media and responds to the actions of

the user. When the user closes the browser, **termination** takes place: the OS stops the process and frees its memory and other resources for other uses.

This lifecycle explains multitasking and concurrency. In **multitasking** the system allows more than one program to be open for a single user, and the user can switch between them, for example listening to music while a document is open. In **concurrency** more than one process is active at the same time, but the CPU processes them one by one in extremely fast cycles, just as a chef prepares three dishes by working on one for a short time and then moving to the next.

3. Differentiate between RAM and virtual memory, and explain how multithreading improves performance.

(a) Difference between RAM and virtual memory:

RAM (primary memory)	Virtual memory
It is the main working area of the computer, made of electronic memory chips.	It is a part of the storage drive, such as a hard drive, an SSD or an NVMe drive, used as extra RAM.
It is very fast, and the CPU can access the data in it quickly.	It is much slower, because the drives have a lower data transfer speed and access rate.
It is limited in size and is quite expensive.	It is much larger in size and cheaper, because the storage drive is big.
Its data is lost when the computer is switched off.	Its data remains safe on the drive, because the drive is non-volatile.
It holds the programs which are currently being used.	It holds the data of the less active programs, which are moved back to RAM when needed.

When the RAM becomes full, the operating system transfers the data of the less active programs to the virtual memory. This frees space in the RAM for the programs which are being used at that time, but the programs kept in virtual memory respond slowly, and the performance of the system is reduced.

(b) How multithreading improves performance: multithreading is a system technique which allows a single process to perform multiple tasks at the same time by dividing its work into smaller units called threads. Each thread runs independently but shares the same memory and resources of its process. It improves performance in the following ways:

(i) Enhanced performance: the tasks are divided into smaller parts and executed in parallel, so complex work is completed more quickly.

(ii) Improved responsiveness: the program remains responsive even when one of its threads is busy in a heavy task. For example, in a word processor one thread keeps accepting your typing while another thread checks the spellings in the background.

(iii) Support for parallel operations: many tasks can be performed in the same period, which is very important in games, video editing and real-time communication tools.

(iv) Efficient use of resources: the threads share the memory space and resources of the parent process, so they need fewer system resources than creating separate processes.

Example: a web browser is one process, but inside it one thread loads and renders the web page, a second thread plays the audio or video, and a third thread downloads a file. Because of this, the browser keeps showing new content while a video is playing, and the user does not have to wait for one task to finish before starting another.

4. Describe system calls, their purpose, and give examples of at least three types.

Definition: a system call is a request made by a program to the operating system to perform a specific task which the program cannot do directly. System calls act as a bridge between the user programs and the kernel.

Purpose: the purpose of a system call is to allow the applications to access the hardware and the core functions of the operating system safely. If every program were allowed to control the hardware directly, the system would become unsafe and unstable, because one faulty program could damage the data of other programs or of the system itself. The system calls also keep the processes separate from one another and maintain the security and stability of the whole system. For example, when you save a file, the program uses a system call to tell the operating system to write the data on the storage drive.

Types of system calls with examples:

System call	What it does	Example
open	Opens a file for reading or writing.	Opening a music file to play it.
read	Retrieves data from a file or from an input device.	Reading the text of a document.
write	Sends data to a file or to an output device.	Saving an image on the computer.
close	Closes a file when its work is finished.	Closing the document after saving it.
fork	Creates a new process.	Opening a new browser tab, which runs as another process.

5. A computer system uses the First-Come, First-Served (FCFS) scheduling method to manage processes. The following table shows the arrival time and the CPU burst time of three processes. Solve the given requirements.

Process	Arrival Time (seconds)	Burst Time (seconds)
P1	0	4
P2	1	3
P3	2	2

I. Order of execution: in FCFS the processes are served in the order in which they arrive, so the order is **P1 → P2 → P3**. P2 and P3 have to wait because they arrive while P1 is still running.

II. Execution sequence:

[P1: 0–4 s] → [P2: 4–7 s] → [P3: 7–9 s]

P1 runs from 0 s to 4 s, then P2 runs from 4 s to 7 s, and finally P3 runs from 7 s to 9 s.

III. Start time and completion time of each process:

Process	Arrival time	Burst time	Start time	Completion time
P1	0 s	4 s	0 s	4 s
P2	1 s	3 s	4 s	7 s
P3	2 s	2 s	7 s	9 s

IV. Waiting time of each process: waiting time = start time − arrival time.

P1 = 0 − 0 = 0 s P2 = 4 − 1 = 3 s P3 = 7 − 2 = 5 s

V. Average waiting time:

Average waiting time = (0 + 3 + 5) ÷ 3 = 8 ÷ 3 = **2.67 seconds**

So all the three processes finish at 9 seconds, the process which had to wait the longest is P3 (5 seconds), and the average waiting time of the system is 2.67 seconds. This example also shows the main drawback of FCFS, because the shortest process P3 was served last of all.